Program : **Diploma in Robotic Process Automation**	
Course Code : **6302B**	Course Title: **Introduction to Computing Problems**
Semester :**6**	Credits: **4**
Course Category: **Program Elective**	
Periods per week: **4 (L: 4 T: 0 P: 0)**	Periods per semester: **60**

Course Objectives:

- To learn basics of digital computer.
- To develop problem solving skills.
- To learn programming and to solve problems using computers.
- To learn programming languages.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course student will be able to:

COn	**Description**	**Duration (Hours)**	**Cognitive Level**
CO1	Ability to design algorithmic solution to problems and to convert algorithms to Python programs.	13	Understanding
CO2	Ability to design modular Python programs using functions	13	Applying
CO3	Ability to design programs with Interactive Input and Output, utilizing arithmetic expression repetitions, decision making, arrays.	19	Understanding & Applying
CO4	Ability to design programs using file Input and Output and to develop recursive solutions.	13	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	3	2	1				2
CO2	3	2	2				2
CO3	3	2	2				2
CO4	3	2	2				2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Ability to design algorithmic solution to problems and to convert algorithms to Python programs.**		
M1.01	Von Neumann concept	1	Understanding
M1.02	A simple model of computer and acquisition of data.	3	Understanding
M1.03	Storage of data and processing of data	2	Understanding
M1.04	Output of processed data.	3	Understanding
M1.05	Details of functional units of a computer.	2	Understanding
M1.06	Storage – primary storage and secondary storage.	2	Understanding
Contents: **Introduction to digital computer –** Von Neumann concept – A simple model of computer, acquisition of data, storage of data, processing of data, output of processed data. Details of functional units of a computer. Storage – primary storage and secondary storage.			
CO2	**Ability to design modular Python programs using functions**		
M2.01	Introduction to programming languages.	1	Understanding.

<table>
<tr><td>M2.02</td><td>Types of programming languages.</td><td>2</td><td>Understanding.</td></tr>
<tr><td>M2.03</td><td>High level language, assembly language and machine language.</td><td>3</td><td>Understanding.</td></tr>
<tr><td>M2.04</td><td>System software</td><td>2</td><td>Understanding.</td></tr>
<tr><td>M2.05</td><td>Operating systems</td><td>2</td><td>Understanding.</td></tr>
<tr><td>M2.06</td><td>Objectives of operating systems, compiler, assembler and interpreter.</td><td>3</td><td>Understanding.</td></tr>
<tr><td></td><td>Series Test – I</td><td>1</td><td></td></tr>
<tr><td colspan="4">Contents:
Introduction to programming languages:- types of programming languages - high level language , assembly language and machine language, System software - Operating systems – objectives of operating systems, compiler, assembler and interpreter.</td></tr>
<tr><td>CO3</td><td colspan="3">Ability to design programs with Interactive Input and Output, utilizing arithmetic expression repetitions, decision making, arrays.</td></tr>
<tr><td>M3.01</td><td>Problem analysis – formal definition of problem</td><td>3</td><td>Applying</td></tr>
<tr><td>M3.02</td><td>Solution – top- down design – breaking a problem into sub problems</td><td>3</td><td>Applying</td></tr>
<tr><td>M3.03</td><td>Overview of the solution to the sub problems by writing step by step procedure (algorithm)</td><td>3</td><td>Applying</td></tr>
<tr><td>M3.04</td><td>Representation of procedure by flowchart</td><td>2</td><td>Applying</td></tr>
<tr><td>M3.05</td><td>Implementation of algorithms – use of procedures to achieve modularity.</td><td>3</td><td>Applying</td></tr>
<tr><td>M3.06</td><td>Implementation of algorithm and flowchart for sample problems.</td><td>2</td><td>Applying</td></tr>
<tr><td>M3.08</td><td>Evaluation of expressions, precedence, string operations.</td><td>3</td><td>Understanding</td></tr>
<tr><td colspan="4">Contents:
Problem Solving strategies –Problem analysis – formal definition of problem – Solution – top- down design – breaking a problem into sub problems- overview of the solution to the sub problems by writing step by step procedure (algorithm) - representation of procedure by flowchart - Implementation of algorithms – use of procedures to achieve modularity.
Introduction to Python – variables, expressions and statements, evaluation of expressions, precedence, string operations</td></tr>
<tr><td>CO4</td><td colspan="3">Ability to design programs using file Input and Output and to develop recursive solutions.</td></tr>
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M4.01	variables, expressions and statements	2	Understanding
M4.02	evaluation of expressions, precedence, string operations	2	Understanding
M4.03	Functions, calling functions, type conversion and coercion	2	Applying
M4.04	Composition of functions, mathematical functions, user-defined functions, parameters and arguments.	2	
M4.05	Strings and lists – string traversal and comparison with examples.	1	
M4.06	Files and exceptions - text files, directories	2	Applying
M4.07	Writing sample recursive programs in python	2	
	Series Test – II	1	

Contents:
Introduction to *Python*: Functions, calling functions, type conversion and coercion, composition of functions, mathematical functions, user-defined functions, parameters and arguments. Strings and lists – string traversal and comparison with examples. Files and exceptions - text files, directories.

Text / Reference

T/R	Book Title/Author
T1	Downey, A. et al., How to think like a Computer Scientist: Learning with Python, John Wiley, 2015
T2	Goel,A., Computer Fundamentals, Pearson Education
T3	Lambert K. A., Fundamentals of Python - First Programs, Cengage Learning India, 2015
T4	Rajaraman, V., Computer Basics and C Programming, Prentice-Hall India

Web Links

T/R	Book Title/Author
1	https://archive.org/details/MIT6.00SCS11
2	https://www.coursera.org/course/pythonlearn